

STELLA Lab Spec parts													
Line	Type	Part	Qty	Category	Type	Vendor	PN	\$ ea	\$ line	main	C2	P2	.STL details, or link to part
15	C2	Cuvette	1	Cuvette config	Consumables	Thomas Scientific	1180W05		0.3	0.3	0	0.3	0
40	P2	PCR 0.2mL tube	1	PCR tube config	Consumables	Fisher Scientific	14-222-262		0.033	0.033	0	0	0.033
1	LS	Adafruit half size proto board	1	Common	Electronics	Adafruit		1609	4.5	4.5	4.5	0	0
2	LS	Magnet coupler 4 pins	1	Common	Electronics	Adafruit		5358	6.5	6.5	6.5	0	0
3	LS	Magnet coupler 3 pins	1	Common	Electronics	Adafruit		5360	6	6	6	0	0
4	LS	Adafruit ADS1015 quad 12 bit ADC	1	Common	Electronics	Adafruit		1083	10	10	10	0	0
5	LS	Adafruit MCP4728 quad 12 bit DAC	1	Common	Electronics	Adafruit		4470	7.5	7.5	7.5	0	0
6	LS	Socket headers, 6 pos	4	Common	Electronics	Digikey	4827-FH254V-06-6TBKG98		0.5	2	2	0	0
7	LS	PN2222ABU transistor	4	Common	Electronics	Digikey	PN2222AFS-ND		0.3	1.2	1.2	0	0
8	LS	1Q 1% 1/4W resistor	4	Common	Electronics	Digikey	13-MF0207FRE52-1RTR		0.1	0.4	0.4	0	0
9	LS	100kQ resistor	4	Common	Electronics	Digikey	RNF14FTD100K		0.1	0.4	0.4	0	0
10	LS	50Q resistor	4	Common	Electronics	Digikey	RNF14FTD49R9		0.1	0.4	0.4	0	0
11	LS	Socket header 3x3 pos	2	Common	Electronics	Digikey	SSW-103-01-G-T		2.6	5.2	5.2	0	0
12	LS	Socket header 3x2 pos	1	Common	Electronics	Digikey	SAM12901-ND		1.5	1.5	1.5	0	0
13	LS	Socket header 3x1 pos	1	Common	Electronics	Digikey	SAM11930-ND		0.7	0.7	0.7	0	0
16	C2	Sparkfun mini proto board	2	Cuvette config	Electronics	Digikey	PRT-12702		4.3	8.6	0	8.6	0
17	C2	Proto-Advantage small proto board	1	Cuvette config	Electronics	Proto-Advantage	SBB1002-1		1	1	0	1	0
18	C2	Adafruit AS7341 spectral sensor	1	Cuvette config	Electronics	Adafruit		4698	19	19	0	19	0
19	C2	Proto-Advantage LED2 - DIP adapter	1	Cuvette config	Electronics	Proto-Advantage	IPC0224		3.5	3.5	0	3.5	0
20	C2	365nm LED 3mm	1	Cuvette config	Electronics	Digikey	1125-MTE3660CS-UV-ND		18	18	0	18	0
21	C2	640nm LED 3mm	1	Cuvette config	Electronics	Digikey	MTE7063C2-UR		2.75	2.75	0	2.75	0
22	C2	488nm LED smt	1	Cuvette config	Electronics	Mouser	GC-VJLPL1.13-KQKS-V2V3-1		1	1	0	1	0
23	C2	Pin header 3x2 pos	1	Cuvette config	Electronics	Digikey	67997-206HLF		0.53	0.53	0	0.53	0
24	C2	Pin header 3 pos	1	Cuvette config	Electronics	Adafruit		392	0.1	0.1	0	0.1	0
25	C2	Pin header single	6	Cuvette config	Electronics	Adafruit		392	0.1	0.6	0	0.6	0
26	C2	500 resistor	2	Cuvette config	Electronics	Digikey	RNF14FTD49R9		0.1	0.2	0	0.2	0
27	C2	Socket header 6 pos	1	Cuvette config	Electronics	Digikey	4627-FH254V-06-6TBKG98		0.5	0.5	0	0.5	0
28	C2	Pin header RA 6 pos	3	Cuvette config	Electronics	Adafruit		1540	0.4	1.2	0	1.2	0
29	C2	White LED panel	1	Cuvette config	Electronics	Adafruit		1626	2	2	0	2	0
41	P2	Sparkfun mini proto board	1	PCR tube config	Electronics	Digikey	PRT-12702		4.3	4.3	0	0	4.3
42	P2	Pin header 3x2 pos	1	PCR tube config	Electronics	Digikey	67997-206HLF		0.53	0.53	0	0	0.53
43	P2	Pin header 3 pos	1	PCR tube config	Electronics	Adafruit		392	0.1	0.1	0	0	0.1
44	P2	Pin header single	12	PCR tube config	Electronics	Adafruit		392	0.1	1.2	0	1.2	0
45	P2	Pin header 6 pos	1	PCR tube config	Electronics	Adafruit		392	0.1	0.1	0	0	0.1
46	P2	Socket header, RA, 6 pos	1	PCR tube config	Electronics	Digikey	S5481-ND		0.7	0.7	0	0	0.7
47	P2	Proto-Advantage small proto board	1	PCR tube config	Electronics	Proto-Advantage	SBB1002-1		1	1	0	0	1
48	P2	Adafruit AS7341 spectral sensor	1	PCR tube config	Electronics	Adafruit		4698	19	19	0	0	19
49	P2	500 resistor	2	PCR tube config	Electronics	Digikey	RNF14FTD49R9		0.1	0.2	0	0	0.2
50	P2	365nm LED 3mm	1	PCR tube config	Electronics	Digikey	1125-MTE3660CS-UV-ND		18	18	0	0	18
51	P2	640nm LED 3mm	1	PCR tube config	Electronics	Digikey	MTE7063C2-UR		2.75	2.75	0	0	2.75
52	P2	488nm LED smt	1	PCR tube config	Electronics	Mouser	720-GCVL113KQKS-V2V31		1	1	0	0	1
53	P2	Proto-Advantage LED2 - DIP adapter	1	PCR tube config	Electronics	Proto-Advantage	IPC0224		3.5	3.5	0	0	3.5
14	LS	LS housing base	1	Common	Print	NA	3d print part		0	0	0	0	0.2mm layers
30	C2	LS-C2 end cover	1	Cuvette config	Print	NA	3d print part		0	0	0	0	0.2mm layers
31	C2	LS-C2 cuvette block	1	Cuvette config	Print	NA	3d print part		0	0	0	0	0.1mm layers
32	C2	LS-C2 cuvette block foot pair, 2up	1	Cuvette config	Print	NA	3d print part		0	0	0	0	0.1mm layers
33	C2	LS-C2 blackout shutter	1	Cuvette config	Print	NA	3d print part		0	0	0	0	0.1mm layers
34	C2	LS-C2 side LED cover, plain	1	Cuvette config	Print	NA	3d print part		0	0	0	0	0.1mm layers
35	C2	LS-C2 side LED cover, flag	1	Cuvette config	Print	NA	3d print part		0	0	0	0	0.1mm layers
36	C2	LS-C2 cuvette pedestal	1	Cuvette config	Print	NA	3d print part		0	0	0	0	0.1mm layers
37	C2	LS-C2 detector block	1	Cuvette config	Print	NA	3d print part		0	0	0	0	0.1mm layers
38	C2	LS-C2 detector block assembly fixture	1	Cuvette config	Print	NA	3d print part		0	0	0	0	0.2mm layers
39	C2	LS-C2 swing lid	1	Cuvette config	Print	NA	3d print part		0	0	0	0	0.2mm layers
54	P2	LS-P2 end cover	1	PCR tube config	Print	NA	3d print part		0	0	0	0	0.2mm layers
55	P2	LS-P2 swing door	1	PCR tube config	Print	NA	3d print part		0	0	0	0	0.1mm layers
56	P2	LS-P2 test position (single or triple source)	1	PCR tube config	Print	NA	3d print part		0	0	0	0	0.1mm layers — Add supports to main overhang, organic style seems to be easier to remove
57	P2	LS-P2 sensor retainer	1	PCR tube config	Print	NA	3d print part		0	0	0	0	0.1mm layers
58	P2	LS-P2 hinge pin	1	PCR tube config	Print	NA	3d print part		0	0	0	0	0.1mm layers
Column cost									158	47	60	53	
C2 Cost for cuvette config plugin									107				
P2 Cost for PCR tube plugin									100				
C2+P2 Cost for set of both									158				

Line	
15	\$30 for 100 pcs
40	\$33 for 1000 pcs
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Table 1

	magnet coupler alignment TBD			
	board holder for magnet coupler TBD			
	smt stencil TBD			
	smt leads TBD			

STELLA Lab Spec Main Board

Step	Action	What value? / Note	Where? / Note	What color?	Done
1	Get board	Adafruit Permaproto Half Size		White	☐
2	Cut traces:		Rear		☐
2.01	Cut trace		30A / 30B		☐
2.02	Cut trace		30B / 30C		☐
2.03	Cut trace		30C / 30D		☐
2.04	Cut trace		30H / 30i		☐
2.05	Cut trace		30i / 30J		☐
2.06	Cut trace		29A / 29B		☐
2.07	Cut trace		29B / 29C		☐
2.08	Cut trace		29C / 29D		☐
2.09	Cut trace		29H / 29i		☐
2.10	Cut trace		29i / 29J		☐
3	Install magnetic couplers:		Front, use fixtures TBD		☐
3.1	Install magnetic coupler	4 pos RA, w spring pins	30: A, B, C, D		☐
3.2	Install magnetic coupler	3 pos RA, w spring pins	30: H, i, J		☐
4	Install bus wires:		Front		☐
4.1	3V		A29-30 to 3V (+ bus below A)	+3.3V	☐
4.2	GND		B29-30 to GND (- bus below A)	GND	☐
4.3	SCL		C29-30 to SCL	i2c Serial Clock	☐
4.4	SDA		E30 to SDA	i2c Serial Data	☐
4.5	5V		J29-30 to G2	5V	☐
5	Install transistors:	PN2222ABU	Front: Flat face towards J: Solder the center pin first, then straighten the body before soldering the other two pins.		☐
5.1	Install transistor	PN2222ABU	B12-13-14		☐
5.2	Install transistor	PN2222ABU	G12-13-14		☐
5.3	Install transistor	PN2222ABU	B16-17-18		☐
5.4	Install transistor	PN2222ABU	G16-17-18		☐
6	Install shunt resistors:	R1Ω 1% tolerance 1/4W	Front		☐
6.1	Install shunt resistor	R1Ω	B21 to GND		☐
6.2	Install shunt resistor	R1Ω	B22 to GND		☐
6.3	Install shunt resistor	R1Ω	B23 to GND		☐
6.4	Install shunt resistor	R1Ω	B24 to GND		☐
7	Install pull down resistors:	100kΩ	Front		☐
7.1	Install pull down resistor	100kΩ	B5 to GND		☐
7.2	Install pull down resistor	100kΩ	B6 to GND		☐
7.3	Install pull down resistor	100kΩ	B7 to GND		☐
7.4	Install pull down resistor	100kΩ	B8 to GND		☐
8	Install bias limit resistors:	R50Ω	Front		☐
8.1	Install bias limit resistor	R50Ω	C11-E13		☐
8.2	Install bias limit resistor	R50Ω	H11-J13		☐
8.3	Install bias limit resistor	R50Ω	C15-E17		☐
8.4	Install bias limit resistor	R50Ω	H15-J17		☐
9	Install socket headers: Front		Front		☐
9.1	Install socket header	6 pos	D4-D9		☐
9.2	Install socket header	6 pos	G4-G9		☐
9.3	Install socket header	6 pos	D20-D25		☐
9.4	Install socket header	6 pos	G20-G25		☐
10	Install socket headers: Rear		Rear		☐
10.1	Install socket header	3x3 (9 pos)	A,B,C x 1,2,3		☐
10.2	Install socket header	3x3 (9 pos)	H,i,J x 1,2,3		☐
10.3	Install socket header	2x3 (6 pos)	GND 1,2,3 and 3V 1,2,3		☐
10.4	Install socket header	1x3 (3 pos)	SDA 1,2,3		☐
11	Install ground wires:		Front	GND	☐
11.1	Install ground wire		B9 to GND	GND	☐

Step	Action	What value? / Note	Where? / Note	What color?	Done
11.2	Install ground wire		F24 to GND	GND	☐
12	Install 3V wires:		Front	+3.3V	☐
12.1	Install 3V wire		B4 to 3V	+3.3V	☐
12.2	Install 3V wire		F25 to 3V	+3.3V	☐
13	Install SCL wires:		Front	i2c Serial Clock	☐
13.1	Install SCL wire		G3 to SCL	i2c Serial Clock	☐
13.2	Install SCL wire		J7 to SCL	i2c Serial Clock	☐
13.3	Install SCL wire		J23 to SCL	i2c Serial Clock	☐
14	Install SDA wires:		Front	i2c Serial Data	☐
14.1	Install SDA wire		J6 to SDA	i2c Serial Data	☐
14.2	Install SDA wire		J22 to SDA	i2c Serial Data	☐
15	Install channel 0 wires:		Front	CH0	☐
15.1	Install channel 0 wire	LED lamp low to collector	F1 to E12	CH0	☐
15.2	Install channel 0 wire	DAC to base through R_limit	C8 to A11	CH0	☐
15.3	Install channel 0 wire	emitter to R_shunt at ADC	A14 to C21	CH0	☐
16	Install channel 1 wires:		Front	CH1	☐
16.1	Install channel 1 wire	LED lamp low to collector	E1 to A16	CH1	☐
16.2	Install channel 1 wire	DAC to base through R_limit	C7 to A15	CH1	☐
16.3	Install channel 1 wire	emitter to R_shunt at ADC	E18 to E22	CH1	☐
17	Install channel 2 wires:		Front	CH2	☐
17.1	Install channel 2 wire	LED lamp low to collector	E2 to F12	CH2	☐
17.2	Install channel 2 wire	DAC to base through R_limit	E6 to F11	CH2	☐
17.3	Install channel 2 wire	emitter to R_shunt at ADC	F14 to E23	CH2	☐
18	Install channel 3 wires:		Front	CH3	☐
18.1	Install channel 3 wire	LED lamp low to collector	E3 to F16	CH3	☐
18.2	Install channel 3 wire	DAC to base through R_limit	E5 to F15	CH3	☐
18.3	Install channel 3 wire	emitter to R_shunt at ADC	F18 to E24	CH3	☐
19	Test board for continuity and shorts				☐
20	Install MCP4728 12 bit DAC		D4-D9; Vcc, VD, VC, VB, VA, GND G4-G9: RDY, LDAC, SDA, SCL, GND, Vcc		☐
21	Install ADS1015 12 bit ADC		D20-D25; A-, A0, A1, A2, A3, A+ G20-G25; ALRT, ADDR, SDA, SCL, GND, Vin		☐
	this page last updated:	20260429			
	reference drawing file:	lab spec main board.PDF			

Line	Dgn	Drawing	PDF
1	LS	LS Main board	yes
2	LS-C2	LS-C2 Cuvette board join	yes
3	LS-C2	LS-C2 Cuvette cantilever board	yes
4	LS-C2	LS-C2 Cuvette cantilever board w fixture	yes
5	LS-C2	LS-C2 Cuvette drop board	yes
6	LS-C2	LS-C2 Cuvette excitation board	yes
7	LS-C2	LS-C2 Cuvette drop and excitation join	yes
8	LS-C2	LS-C2 Cuvette optics	yes
9	LS-C2	LS-C2 Cuvette optics	TBD
10	LS-C2	LS-C2 Assembly drawing	TBD
11	LS-P2	LS-P2 PCR tube cantilever board	yes
12	LS-P2	LS-P2 PCR tube excitation board	yes
13	LS-P2	LS-P2 sensor block board set	yes
14	LS-P2	LS-P2 PCR tube optics	yes
15	LS-P2	LS-P2 Assembly drawing	TBD

surface mount technology LED adapter board build

Step				
1	print fixtures			
2	clean adapter board and stencil w isopropanol			
3	identify and mark LED (-)			
4	insert adapter board in stencil fixture per diagram			
5	insert stencil in stencil fixture per diagram			
6	extrude a small amount of solder paste across the ends of the stencil slots, like the upper cross bar on a π symbol.			
7	use the squeegee to fill the stencil slots with solder paste			
8	carefully remove the stencil without disturbing the = made by the solder paste.			
9	use fine tweezers to orient the LED correctly: The dot you make in 3 should point to the pin 1 side of the board, where there is a little white dot to match.			
10	place the LED on the solder paste =. It doesn't have to be exactly aligned, the surface tension will pull it into alignment when the solder is molten.			
11	heat the hotplate to TBD°C BE VERY CAUTIOUS, it's a very hot stove. Don't let any part of your hand make contact with the hot surface.			
12	carefully extract the board, with the LED balanced on the solder paste, from the stencil fixture			
13	...heat, watch for melt CAREFULLY			
14	...remove from heat. allow to cool.			
15	...probe w meter set to diode mode to check polarity.	if incorrect, install backwards on excitation board		
16	...set board, diode down, in the pin alignment fixture			
17	... set pins on board, using pin alignment board...			
18	... solder outer pins			
19	... remove alignment board			
20	... solder inner pins			
21	... check polarity again.			
22	... install on excitation board			

STELLA Lab Spec Cuvette Cantilever Board

Step	Action	What value?	Where?	What color?	Done
1	Get board	Sparkfun mini		Red	☒
2	Using main board as a template, install pin headers	3x2 one place, 3 pos one place, 1 pos six places	1-2ABC, A4, B5, C6, C13, B14, A16, 17ABC		☒
3	Fixture detector board at right angles to cantilever board	6 pos right angle pin header	G7-12		☒
4	Install socket header for drop board	6 pos socket header	G1-6		☒
5	<i>Note: flatten the wiring along row F to clear the drop board</i>	<i>Wires can stick up between D, E and F</i>			☐
6	Install ground wires		E2-F11	GND	☒
7	Install power wires		E1-F12	+3.3V	☒
8	Install SCL wires		E15-F10	i2c Serial Clock	☒
9	Install SDA wires		E17-F9	i2c Serial Data	☒
10	Install 5V wire		E14-F6	5V	☒
11	Install channel 0 wires		E13-F5	CH0	☒
12	Install channel 1 wires		E6-F4	CH1	☒
13	Install channel 2 wires		E5-F3	CH2	☒
14	Install channel 3 wires		E4-F2	CH3	☒
15	Test fit the drop board				☒
16	Test board for continuity and shorts				☐

STELLA Lab Spec Cuvette Drop Board

Step	Action	What value?	Where?	What color?	Done
1	Get board	Proto-Advantage 1.1x1.7"		Epoxy color	☐
2	Cut traces		F1-2, F2-3, F3-4, F4-5, F5-6 F6-7		☐
3	install right angle pin headers, long pins towards the edges of the board	6 pos, two places	B1-6, G1-6		☐
4	Install current limit resistors, short to G pins on header	50Ω, three places	FG2-C2, FG3-C3, FG4-C4		☐
5	Install channel 0 wires, short to G pin on header		FG5-C5	CH0	☐
7	Install 5V wire, short to G pin on header		FG6-C6	5V	☐
7	Test board for continuity and shorts				☐

STELLA Lab Spec Cuvette Excitation Board

Step	Action	What value?	Where?	What color?
1	Get board	Sparkfun mini		Red
2	Cut traces		F1-G1, F16-G16	
3	Install bottom LED, smt	488nm	(1,dot) FG7, (2,+) FG10	
4	Trim backlight LED panel (score with diagonal cutters, snap, trim corners)	Shorten the panel by 15mm	TBD fixture for this step	
5	Install CH3 backlight LED panel		(-) J8, (+) J9	
6	Bend CH1 and CH2 LEDs, match polarity and height below			
7	Install CH1 LED, set height with cuvette block	365nm (for DAPI)	(+) F1, (-) G1	
8	Install CH2 LED, set height with cuvette block	640nm (for Cy5)	(+) F16 (-) G16	
9	Install 5V wires		i13 - i10+i9, C12 - F13, H13 - F15+F16, D12 - F1+F2	5V
10	Install channel 0 wires		D13 - H7	CH0
11	Install channel 1 wires		F14 - i1	CH1
12	Install channel 2 wires		F15- i16	CH2
13	Install channel 3 wires		F16 - H8	CH3
	join to drop board... TBD			
14	Test board for continuity and shorts			

	pull 6 pins from a spare header			
	use the remaining plastic bar to double the spacer off the A side of the drop board			
	connect the boards at A12 on the cantilever board, A17 on the excitation board			

STELLA Lab Spec PCR Tube Cantilever Board

Step	Action	What value?	Where?	What color?
1	Get board	Sparkfun mini		Red
2	Cut traces		GH7, ...GH12	
3	Pull 6 pins from a spare pin header			
4	Install the pins, sticking up from the board, into a right angle socket header		F7,...F12	
5	Using main board as a template, install pin headers	3 pos in three places, 1 pos in six places	1ABC, A4, B5, C6, C13, B14, A16, 17ABC	
6	Install ground wires	REAR, single layer, no crossovers	H8 - E2	GND
7	Install power wires	REAR, single layer, no crossovers	H7 - E1	+3.3V
8	Install SCL wires	REAR, single layer, no crossovers	H9 - E13	i2c Serial Clock
9	Install SDA wires	REAR, single layer, no crossovers	i10 - E17	i2c Serial Data
10	Install 5V wire	REAR, single layer, no crossovers	G10 - D14	5V
11	Install channel 0 wires	REAR, single layer, no crossovers	G11 - E15	CH0
12	Install channel 1 wires	REAR, single layer, no crossovers	G12 - E6	CH1
13	Install channel 2 wires	REAR, single layer, no crossovers	G8 - E5	CH2
14	test the sensor function with the qwiic connect cable before installing			
15	Mount the detector board, face up, extending off the cantilever board		J7, ...J12	
16	Test board for continuity and shorts			

STELLA Lab Spec PCR tube excitation board

Step	Action	What value?	Where?	What color?
1	Get board	Proto-Advantage 1.1x1.7"		Epoxy color
2	Cut trace		E6-E7	
3	install right angle pin headers	6 pos, 2 places	i4, ... i9	
4	test the CH0 LED polarity before installing			
5	install CH0 LED		[+,2] G6,G7 [-,1, dot] D6,D7	
6	install CH1 LED		[+] F11, [-] E11	
7	install CH2 LED		[+] F2, [-] E2	
8	Install channel 0 wires		REAR: G8 - C7	CH0
9	Install current limit resistor CH1	50Ω	REAR: G9 - E10	CH1
10	Install current limit resistor CH2	50Ω	REAR: G4 - G1	CH2
11	Test board for continuity and shorts			

Table 1

	units	value		
hfe ~ gain PN2222		100		
max current	mA	700		
base current	mA	7		
base resistor	Ω	50		
BE voltage	V	0.7		
DAC max	V	1.05		
Supply	V	3.3		
DAC max / Supply		0.3		
DAC set max		65535		
DAC use max		19660.5		
DAC nominal max		20000		

Line	condition	program action	result	note	result
0	INTERFACE TESTS:				
1	date correct		PASS		
2	time correct and live		PASS		
3	selection change on rotary		PASS		
4	correct selection order		PASS		
5	push when main selected	go to main page	PASS		
6	push when B+ selected	increment batch number	PASS		
7	batch number correct		PASS		
8	battery % correct		PASS		
9	status correct		PASS		
10	status highlight color correct		PASS		
11	measure button color correct		PASS		
12	push when measure selected	run measurement sequence	PASS		
13	push when excitation selected	selectable, green	PASS		
14	second push when excitation selected	not selectable, white	PASS		
15	rotate when excitation selectable	cycle lamp choice	PASS	make this list wrap around	DONE
16	excitation lamp ID correct		PASS		
17	push when current selected	selectable, green, show REQ	PASS		
18	second push when current selected	not selectable, white, show last	PASS		
19	rotate when current selectable	change REQ %	PASS		
20	current percentage or last correct		PASS		
21	push when gain selected	selectable, green	PASS		
22	second push when gain selected	not selectable, white	PASS		
23	rotate when gain selectable	change gain	PASS		
24	gain correct		PASS		
25	push when integration selected	selectable, green	PASS		
26	second push when integration selected	not selectable, white	PASS		
27	rotate when integration selectable	change integration	PASS		
28	integration correct		PASS		
29	push when view A selected	selectable, green	PASS		
30	second push when view A selected	not selectable, white	PASS		
31	rotate when view A selectable	change view A channel	PASS		
32	view A correct		PASS		
33	push when view B selected	selectable, green	PASS		
34	second push when view B selected	not selectable, white	PASS		
35	rotate when view B selectable	change view B channel	PASS		
36	view B correct		PASS		
37	show A B choice collision warning colors		PASS		
38	instruction line		TBD	not implemented yet	
39	back arrow		TBD	not implemented yet	
40	DO		TBD	not implemented yet	
41	next arrow		TBD	not implemented yet	
42	M number correct		PASS		
43	A value correct		PASS		
44	B value correct		PASS		
45	Ratio correct		PASS		
46	%DR correct		PASS		
47	make mmt display values change with A B choice		TBD	not implemented yet	
100	FUNCTIONAL TESTS:				
101	read current on lamp choice ADC channel		PASS		
	assign correct part numbers to each lamp				

Line	condition	program action	result	note	result
1000	ADD FUNCTIONALITY:				
1001	work with as7343 sensor				
1002	accept configuration file to describe lamps				